

# SuperAI SS6

\$0 offline

35/35 tests

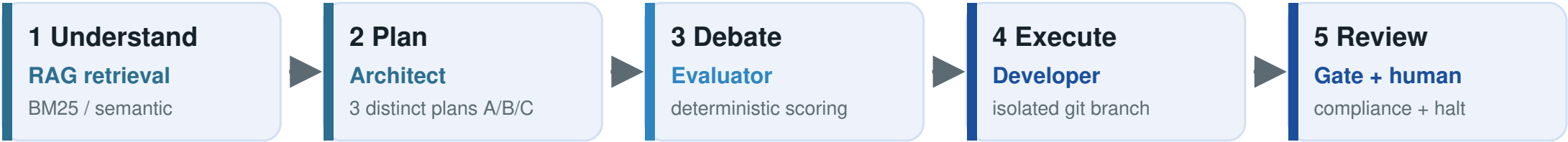
eleven-7 case study

## An Autonomous AI Software-Engineering Pipeline

A plain-English requirement → a conventions-compliant, test-passing change on an isolated branch — with the decision made by auditable math and the merge reserved for a human.

### The pipeline — five roles, like a small engineering team

Orchestrated as a LangGraph state machine; every stage has its own evaluation harness.



#### The repair loop (what makes it converge)

generate → gate → feed violations back → regenerate



##### Offline demo (deterministic, \$0):

Attempt 1 direct fetch() — context.md §4  
Attempt 2 auto-fixed → gate PASS

#### Safety invariants (never traded away)

- Hard repo isolation — all work in a throwaway copy
- Deterministic, explainable Debate — reproducible winner
- Human-in-the-loop — pipeline halts, never auto-merges
- An eval harness per phase — measured, not asserted

#### Results

35 / 35

unit tests green

\$0

fully offline (mock)

tsc + jest

real gate, isolated copy

Recall@5 89%

BM25, +39 pts vs old

#### Retriever baseline (18-query set)

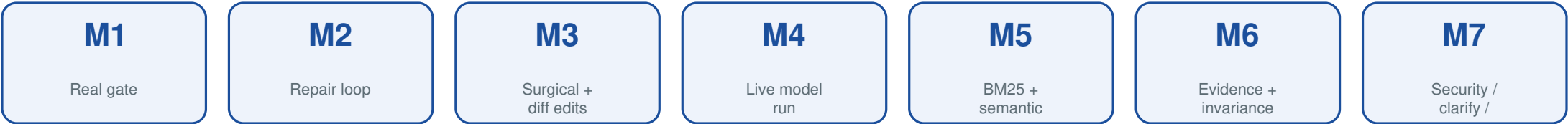
|                        | R@1       | R@3       | R@5       | FP        |
|------------------------|-----------|-----------|-----------|-----------|
| hashing (old)          | 33        | 44        | 50        | 50        |
| <b>BM25 + coverage</b> | <b>78</b> | <b>83</b> | <b>89</b> | <b>25</b> |
| <b>semantic MiniLM</b> | <b>72</b> | <b>89</b> | <b>94</b> | <b>75</b> |

BM25 wins R@1 & false-positives; semantic wins depth → hybrid next.

### Live run on Ollama qwen2.5-coder:7B — the gate has teeth

The real model's code compiled and passed jest (tsc jest) — but violated context.md (didn't reuse the canonical primitives; touched a shared one). The repair loop ran all 3 attempts; **the model couldn't converge → gate stayed FAIL → pipeline HALTED. Non-compliant code never reached review.**

### Roadmap status



### Why it matters for real software engineering

#### Convention enforcement at scale

team rules encoded once; every change auto-checked before a human looks.

#### Drafts become reviewable PRs

a requirement lands as a compliant diff on a branch, not a blank file.

#### Trust by construction

the consequential decision is auditable math; the merge stays human.